COMP3670/6670
Introduction to Machine Learning
Week 6b: Span, Independence, Bases, Dimension & Rank

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Goals of this lecture

- Interpret predictions using the column space of a data matrix.
- Use the null space to describe changes that leave predictions unchanged.
- Identify linearly independent spanning sets, bases and dimensions.
- Use pivots to compute rank and nullity, then interpret rank in ML.

- A data matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ stores one feature in each column.
- If its columns are $\mathbf{c}_1, \ldots, \mathbf{c}_d$, then

$$\mathbf{X}\mathbf{w} = w_1\mathbf{c}_1 + \cdots + w_d\mathbf{c}_d.$$

- Which prediction vectors can the model produce?
- Which features provide genuinely new directions?
- What is the smallest number of feature columns that preserves every possible prediction?

Linear combinations describe available predictions

Example 1. Four datapoints are described by four features:

$$\mathbf{X} = \begin{pmatrix} 3 & 4 & 7 & 5 \\ 5 & 7 & 12 & 9 \\ 8 & 11 & 19 & 14 \\ 6 & 9 & 15 & 8 \end{pmatrix} = \begin{pmatrix} | & | & | & | \\ \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 & \mathbf{c}_4 \\ | & | & | & | \end{pmatrix}.$$

For

$$\mathbf{w} = \begin{pmatrix} 8 \\ -3 \\ 0 \\ 6 \end{pmatrix},$$

what prediction vector does the model produce?

Example 1 (cont'd). The prediction vector is

$$\mathbf{X}\mathbf{w} = 8\mathbf{c}_1 - 3\mathbf{c}_2 + 6\mathbf{c}_4 = \begin{pmatrix} 42 \\ 73 \\ 115 \\ 69 \end{pmatrix}.$$

The model scales and adds feature columns. Moreover,

$$\mathbf{c}_3 = \mathbf{c}_1 + \mathbf{c}_2.$$

Thus the third feature contributes no direction that the first two features cannot already provide.

Linear combinations and span. A linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^m$ has the form

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k, \qquad \alpha_1, \ldots, \alpha_k \in \mathbb{R}.$$

Their span is the set of all such vectors:

$$\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} = \left\{ \sum_{j=1}^{k} \alpha_j \mathbf{v}_j : \alpha_j \in \mathbb{R} \right\}.$$

If $\mathbf{V} = \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_k \end{pmatrix}$, this is $\{\mathbf{V}\boldsymbol{\alpha} : \boldsymbol{\alpha} \in \mathbb{R}^k\}$.

Subspace. A subset $\mathcal{S} \subseteq \mathbb{R}^m$ is a subspace if

1. $\mathbf{0} \in \mathcal{S}$;
2. $\mathbf{x} + \mathbf{y} \in \mathcal{S}$ whenever $\mathbf{x}, \mathbf{y} \in \mathcal{S}$;
3. $a\mathbf{x} \in \mathcal{S}$ whenever $\mathbf{x} \in \mathcal{S}$ and $a \in \mathbb{R}$.

A subspace contains the zero vector and is closed under addition and scaling.

Proposition 1. For any $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^m$,

$$\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$$

is a subspace of $\mathbb{R}^m$.

It is the smallest subspace containing every vector $\mathbf{v}_1, \ldots, \mathbf{v}_k$.

Proof of Proposition 1.

Column and null spaces

Column and null spaces. Let $\mathbf{A} \in \mathbb{R}^{m \times d}$ have columns $\mathbf{a}_1, \ldots, \mathbf{a}_d$. Its column space is

$$\mathrm{Col}(\mathbf{A}) = \mathrm{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_d\} = \{\mathbf{A}\mathbf{w} : \mathbf{w} \in \mathbb{R}^d\} \subseteq \mathbb{R}^m.$$

Its null space is

$$\mathrm{Nul}(\mathbf{A}) = \{\mathbf{z} \in \mathbb{R}^d : \mathbf{A}\mathbf{z} = \mathbf{0}\} \subseteq \mathbb{R}^d.$$

The column space lives in the output space; the null space lives in the input space.

Proposition 2. For every matrix $\mathbf{A}$, both

$$\mathrm{Col}(\mathbf{A}) \qquad \text{and} \qquad \mathrm{Nul}(\mathbf{A})$$

are subspaces of their respective ambient spaces.

Proof of Proposition 2 (for $\mathrm{Col}(\mathbf{A})$).

Proposition 3. Let $\mathbf{A} \in \mathbb{R}^{m \times d}$ and $\mathbf{b} \in \mathbb{R}^m$.

- The system $\mathbf{A}\mathbf{w} = \mathbf{b}$ is consistent exactly when $\mathbf{b} \in \mathrm{Col}(\mathbf{A})$.
- If $\mathbf{w}_0$ is one solution, then every solution is obtained as

$$\mathbf{w}_0 + \mathbf{z}, \qquad \mathbf{z} \in \mathrm{Nul}(\mathbf{A}).$$

  Equivalently, the solution set is $\mathbf{w}_0 + \mathrm{Nul}(\mathbf{A})$.

Linear independence, bases and dimension

Linear independence. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are linearly independent if

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0}$$

implies

$$\alpha_1 = \cdots = \alpha_k = 0.$$

Otherwise, they are linearly dependent.

For a data matrix, dependent columns indicate redundant feature directions and weight vectors that cannot be distinguished by their predictions.

Example 2. Consider

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}, \qquad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}.$$

Suppose $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = \mathbf{0}$. Expanding:

$$\begin{pmatrix} \alpha_1 + \alpha_2 + 2\alpha_3 \\ 4\alpha_2 \\ 2\alpha_2 + 5\alpha_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

which implies $\alpha_2 = 0$, then $\alpha_3 = 0$, then $\alpha_1 = 0$. Thus, the three vectors are linearly independent.

Example 2 (cont'd). In Example 1, the feature columns satisfy

$$\mathbf{c}_3 = \mathbf{c}_1 + \mathbf{c}_2.$$

Equivalently,

$$-\mathbf{c}_1 - \mathbf{c}_2 + \mathbf{c}_3 = \mathbf{0}.$$

The coefficients $-1, -1, 1$ are not all zero. Hence $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ are linearly dependent.

Basis. Let $\mathcal{S}$ be a subspace. A list $\mathbf{v}_1, \ldots, \mathbf{v}_r \in \mathcal{S}$ is a basis of $\mathcal{S}$ if

- $\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_r\} = \mathcal{S}$;
- $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are linearly independent.

If $\mathbf{x} = \alpha_1 \mathbf{v}_1 + \cdots + \alpha_r \mathbf{v}_r$, then $(\alpha_1, \ldots, \alpha_r)^{\mathsf{T}}$ is the coordinate vector of $\mathbf{x}$ relative to this ordered basis.

Proposition 4. Every vector in a subspace has exactly one coordinate vector relative to an ordered basis of that subspace.

Theorem 5. Suppose $\mathbf{v}_1, \ldots, \mathbf{v}_r$ span a subspace $\mathcal{S}$, while $\mathbf{u}_1, \ldots, \mathbf{u}_k \in \mathcal{S}$ are linearly independent. Then

$$k \leq r.$$

Consequently:

- any two bases of $\mathcal{S}$ have the same number of vectors;
- every subspace of $\mathbb{R}^m$ has a basis containing at most m vectors.

Proof of Theorem 5. Let

$$\mathbf{V} = \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_r \end{pmatrix}, \qquad \mathbf{U} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k \end{pmatrix}.$$

Since the columns of $\mathbf{U}$ lie in the span of the columns of $\mathbf{V}$, there is a matrix $\mathbf{C} \in \mathbb{R}^{r \times k}$ such that

$$\mathbf{U} = \mathbf{V}\mathbf{C}.$$

If $k > r$, then $\mathbf{C}\mathbf{z} = \mathbf{0}$ has a nonzero solution. Hence

$$\mathbf{U}\mathbf{z} = \mathbf{V}\mathbf{C}\mathbf{z} = \mathbf{0},$$

contradicting the independence of the columns of $\mathbf{U}$. Therefore $k \leq r$.

Dimension. The dimension of a subspace $\mathcal{S}$, denoted $\dim(\mathcal{S})$, is the number of vectors in any basis of $\mathcal{S}$.

Theorem 5 guarantees that this number does not depend on which basis we choose.

For example, the standard vectors $\mathbf{e}_1, \ldots, \mathbf{e}_m$ form a basis of $\mathbb{R}^m$, so

$$\dim(\mathbb{R}^m) = m.$$

Rank counts usable directions

- So far, we have learnt that any subspace has a dimension, which is the minimum number of linearly independent vectors that span the subspace.
- Column space $\mathrm{Col}(\mathbf{A})$ and null space $\mathrm{Nul}(\mathbf{A})$ are subspaces, so they have dimensions.

Rank and nullity. For $\mathbf{A} \in \mathbb{R}^{m \times d}$,

$$\operatorname{rank}(\mathbf{A}) = \dim(\operatorname{Col}(\mathbf{A}))$$

and

$$\operatorname{nullity}(\mathbf{A}) = \dim(\operatorname{Nul}(\mathbf{A})).$$

Rank counts independent output directions; nullity counts independent input directions that the matrix sends to zero.

Theorem 6 (Rank-nullity). Let $\mathbf{A} \in \mathbb{R}^{m \times d}$, and suppose its row-echelon form has r pivots. Then:

- the pivot columns of the original matrix form a basis of $\mathrm{Col}(\mathbf{A})$;
- $\mathrm{rank}(\mathbf{A}) = r$;
- $\mathrm{nullity}(\mathbf{A}) = d - r$.

Therefore,

$$\mathrm{rank}(\mathbf{A}) + \mathrm{nullity}(\mathbf{A}) = d.$$

Corollary 7. If $\mathbf{A}$ has rank r, then r is the minimum number of columns of $\mathbf{A}$ whose span equals $\mathrm{Col}(\mathbf{A})$.

In particular, the pivot columns form such a minimum collection.

For a data matrix, rank is the minimum number of original feature columns needed to preserve every prediction the linear model can produce.

Example 3. The rows of this data matrix represent ten datapoints in $\mathbb{R}^7$:

$$\mathbf{X} = \begin{pmatrix} 3 & 9 & 4 & -6 & -4 & 5 & 15 \\ 17 & 51 & 26 & -44 & -16 & 23 & 71 \\ 6 & 18 & 9 & -15 & -6 & 8 & 25 \\ 15 & 45 & 20 & -30 & -20 & 29 & 83 \\ 5 & 15 & 7 & -11 & -6 & 9 & 26 \\ 11 & 33 & 16 & -26 & -12 & 13 & 43 \\ 5 & 15 & 5 & -5 & -10 & 13 & 36 \\ 15 & 45 & 23 & -39 & -14 & 15 & 52 \\ 25 & 75 & 38 & -64 & -24 & 35 & 107 \\ 13 & 39 & 20 & -34 & -12 & 11 & 41 \end{pmatrix} \in \mathbb{R}^{10\times 7}.$$

Example 3 (cont'd). Gaussian elimination gives

$$\mathbf{X} \sim \begin{pmatrix} 3 & 9 & 4 & -6 & -4 & 5 & 15 \\ 0 & 0 & 1 & -3 & 2 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 & -4 & -8 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivots lie in columns 1, 3 and 6.

Example 3 (cont'd). The pivot columns 1, 3 and 6 of the original matrix form a basis of $\mathrm{Col}(\mathbf{X})$. Hence

$$\mathrm{rank}(\mathbf{X}) = 3.$$

Row operations also preserve the span of the rows, and the three nonzero echelon rows are independent because their pivots occur in different positions.

Thus the row space has dimension 3. Although the dataset contains ten datapoints in $\mathbb{R}^7$, all ten lie in a subspace of dimension only 3.

- If datapoints lie in an r-dimensional subspace of $\mathbb{R}^d$ with $r < d$, they have only r effective degrees of freedom.
- Each datapoint can then be represented by r coefficients rather than d, which often also reduces the number of effective parameters that must be learned.
- This motivates dimensionality reduction. PCA will later seek a low-dimensional subspace that preserves as much variation as possible.

Takeaways

- Span records every vector obtainable by linear combination.
- Column space describes attainable outputs; null space describes invisible input changes.
- A basis spans without redundancy, and its size is the dimension.
- Pivot columns give a basis of the column space.
- Rank counts usable directions.